

# Pei-Dong Li

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## Research Interests

Trapped-ion quantum information; surface-electrode ion traps; quantum control and readout of motional degrees of freedom in single ions; single-ion quantum thermodynamics and quantum heat engines; coherent spin-motion dynamics, laser cooling.

## Education

**Ph.D. in Physics**, Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences (APM, CAS), Wuhan, China 2021–Winter 2026

- Ph.D. defence (expected): Winter 2026
- Division of Atomic, Molecular and Optical Physics
- Advisor: Prof. Mang Feng (Researcher, Group PI)
- Co-supervisor: Dr. Liang Chen (Associate Researcher)
- Research focus: Trapped-ion quantum thermodynamics and motional-state engineering

**B.Sc. in Physics**, Wuhan University, Wuhan, China 2017–2021

## Research Experience

**Single-ion phonon laser in the quantum regime** 2025–2026

- Lead experimentalist for single-ion phonon lasing, developing experimental scheme and realizing phonon-lasing states.
- Exploring applications for weak AC electric field sensing using phonon-lasing states.

**Hybrid exceptional points in dissipative quantum systems** 2025

- Led experimental implementation and data acquisition for a non-Hermitian system.
- Constructed hybrid exceptional points and observed second- and third-order eigenenergy splittings.
- Demonstrated enhanced response near higher-order exceptional points, relevant for quantum sensing.

**Single-ion Quantum Szilárd Engine** 2024

- Led experimental implementation on a surface-electrode ion trap and established motional-state control.
- Designed measurement protocols and performed motional-state reconstruction.
- Prepared figures and contributed to manuscript writing.

**Quantum jump induced Liouvillian exceptional points** 2024

- Led experimental setup, calibration, and execution on a trapped-ion platform.
- Assisted in selection of experimental parameters and performed data acquisition.
- Wrote the experimental section of the manuscript.

**Single-spin Quantum Stirling Engine** 2022–2023

- Led experimental implementation and data acquisition.
- Assisted in experimental design and performed data analysis.

## Single-ion experiments on a surface-electrode trap

2021–present

- Setup, optimized, and maintained the surface-electrode ion trap system.
- Performed micromotion compensation and sideband cooling to near motional ground state.
- Established coherent control of both internal (spin) and external (motional) degrees of freedom.

## Technical Skills

**Experimental:** Surface-electrode ion trap operation, micromotion compensation, sideband cooling, coherent spin-motion control, Fock-state preparation and phonon-number reconstruction, experimental control and calibration.

**Computational:** Python, MATLAB, LabVIEW, SolidWorks, Altium Designer; numerical simulation and analysis of trapped-ion systems.

## Awards & Honors

- Outstanding Student Leader, University of Chinese Academy of Sciences 2021–2022, 2022–2023
- Merit Student, University of Chinese Academy of Sciences 2023–2024, 2024–2025, 2025–2026
- Third Prize, Wuhan City Science Popularization Contest 2024
- Excellence Award, Hubei Province Science Popularization Short Video Contest (Social Group) 2024

*“From Indivisible to Quantum Miracle: Journey of Atoms”*

## Publications

- **Pei-Dong Li**, Yuan-Zhang Dong, Zhuo-Zhu Wu, et al., “Finite-time electrometry with a quantum-regime single-ion phonon laser,” Under submission (2026).
- **Pei-Dong Li\***, Zhuo-Zhu Wu\*, et al., “Continuous activation of quantum jumps bridges Hamiltonian and Liouvillian exceptional points,” Under submission (2026).
- **Pei-Dong Li\***, Zhuo-Zhu Wu\*, Kai-Feng Cui\*, et al., “Experimental witness of phonon-transfer trajectories in a cyclic quantum Szilárd engine,” Under submission (2026).
- Zhuo-Zhu Wu\*, **Pei-Dong Li\***, Tai-Hao Cui, et al., “Experimental witness of quantum jump induced high-order Liouvillian exceptional points,” *Nature Communications* **17**, 1923 (2026). DOI
- **Pei-Dong Li\***, G.-Y. Ding\*, J.-Q. Zhang, et al., “Experimental demonstration of single-spin Stirling engine cycles with enhanced efficiency,” *Physical Review A* **111**, L010203 (2025). DOI

*\*Co-first authors contributed equally.*

## Referees

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